Roll No. | | | | | | | |

END SEMESTER EXAMINATION DECEMBER- 2025

Semester: 5th

Name of the Course: B. Tech (CSE)

Name of the Paper: **Introduction to Artificial Intelligence and Data Science** Paper Code: TCS-562

Time: 3 Hours Maximum Marks: 100

Note:-

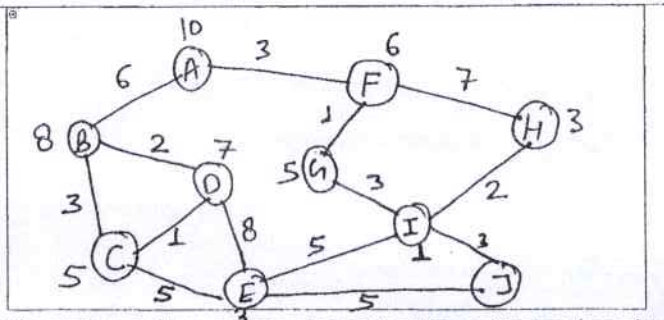
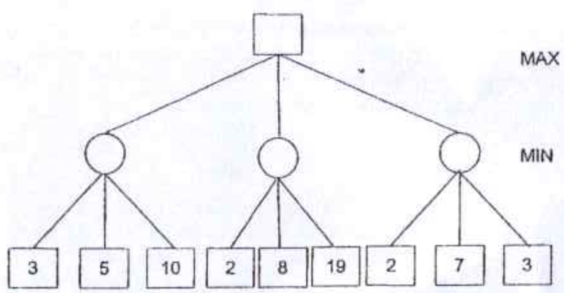
(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b and c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

		CO
Q1	(20 marks)	
(a)	Definition and evolution of Data Science. Explain the key terminologies of data, information, knowledge, analytics, big data, etc. Discuss the Data Science process model (CRISP-DM framework) and their importance in solving real-world problems	2
(b)	1. The different categories of AI systems (Reactive Machines, Limited Memory, Theory of Mind, Self-Aware Systems). (4marks) 2. Real-world applications of each type in areas such as healthcare, education, robotics, finance, and transportation. (3 marks) 3. A discussion on current trends in AI, such as generative AI, autonomous systems, and ethical AI. (3 marks)	2
©	Compare the applications and processes of Artificial Intelligence and Data Science in healthcare and finance. How do current trends in AI enhance data-driven decision-making in these domains?	CO1
Q2	(20 marks)	
(a)	1. Definitions and formulas of mean, median, and mode. (3 marks) 2. Measures of dispersion such as range, variance, standard deviation, and interquartile range. (4 marks) 3. Appropriate examples or graphs illustrating normal and non-normal distributions. (3 marks)	3,4,5
(b)	1. Steps in understanding data requirements and identifying independent and dependent variables. (3 marks) 2. Methods for handling missing or erroneous data. (3 marks) 4.Python-based examples or code snippets for data cleaning, visualization, and summary statistics generation. (4 marks)	
©	1. Explanation of null and alternative hypotheses, p-value, and level of significance. (3 marks) 2. The interpretation of Root Mean Square Error (RMSE) as a measure of model accuracy. (3 marks) 3. Real-world examples illustrating how these tests support decision-making and model evaluation. (4 marks)	
Q3	(20 marks)	
(a)	Perform K-Means Clustering, Given three data points: A(2, 4), B(3, 5), C(8, 9). Assume two cluster centroids are initially: $C_1(2, 3)$ and $C_2(7, 8)$,Tasks: 1. Compute the Euclidean distance of each point from both centroids. (4 marks) 2. Assign each data point to the nearest cluster. (3 marks) 3. Calculate the new cluster centroids after the first iteration. (3 marks)	3,4,5
(b)	The following transaction dataset contains 5 transactions from a small store: Given: Minimum Support = 60%, and Minimum Confidence = 80% **Transaction ID** / **Items Purchased**: **T1** — **{Bread, Milk}** **T2** — **{Bread, Diaper, Beer, Eggs}** **T3** — **{Milk, Diaper, Beer, Coke}** **T4** — **{Bread, Milk, Diaper, Beer}** **T5** — **{Bread, Milk, Diaper, Coke}** Tasks: 1. Find all frequent item sets using the Apriori algorithm (L1, L2, L3) based on the given support threshold. *(6 marks)* 2. From the frequent item sets, generate strong association rules that satisfy the minimum confidence level. *(3 marks)* 3. Interpret one of the generated rules in practical terms (e.g., what it means for sales prediction or recommendation). *(1 mark)*	
(c)	Dimensionality Reduction and Outlier Detection given the following 2-dimensional dataset: **Point X_1 X_2** 1 2 3 2 3 3 3 4 4 4 5 5	

	5 100 105 Tasks: 1. Compute the mean and standard deviation of each variable. (4 marks) 2. Identify if there are outliers using the Z-score method (use threshold = 3). (4 marks) 3. Discuss briefly how PCA (Principal Component Analysis) could help visualize this dataset in lower dimensions. (2 marks)	
Q4	(20 marks)	
(a)	 The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.	2,4,3
(b)	1. Apply the Alpha-Beta pruning algorithm to the given tree. 2. Show the updates of α (alpha) and β (beta) values at each node. 3. Identify which branches are pruned. 4. State the final minimax value returned to the root node (A). 	

(c)	Explain the concept of Knowledge Representation and Reasoning in Artificial Intelligence. Discuss Knowledge-Based Agents – their architecture and working principles.	
Q5	(20 marks)	
(a)	Describe the structure and working of an Expert System (MYCIN, DENDRAL): • Definition and role of Expert Systems in Artificial Intelligence. (3marks) • Explanation of Expert System Architecture (Knowledge Base, Inference Engine, User Interface, Explanation Facility) (4marks) • The process of reasoning using rules (if-then statements) (3marks)	5
(b)	Explain the Applications of Analytics in Healthcare, Business domains. Include relevant examples such as: 1. Predictive models in healthcare for disease diagnosis. (5marks) Customer analytics in business for marketing optimization. (5marks)	6
(c)	Evaluate the impact of Analytics and Expert Systems through real-world case studies. Choose any two case studies from different domains and explain: 1. The problem scenario and data involved 2. The analytical or expert system model applied 3. The insights, decisions, or outcomes achieved 4. Challenges and limitations faced in implementation	6